

Missing Creatures Implementation Plan

D&D 3.5e NPCDatabase — Adding 74 Missing Creatures

Project: dnd35prototype

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Current Coverage: 124/198 creatures (62.6%)

Target Coverage: 198/198 creatures (100%)

Missing Creatures: 74

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Overview

Current State

The NPCDatabase has strong coverage of:

- **Dragons:** 60 variants (all age categories × 10 types)
- **Mephits:** All 10 types
- **Vermin:** Size-scaled centipedes (7), scorpions (3), spiders (3)
- **Templates:** 9 creature templates (Skeleton, Zombie, Lycanthrope variants, Celestial/Fiendish)
- **Classes:** 16 character classes via CreatureClassEngine
- **Spawner:** DungeonEncounterSpawner with dynamic class levels and template application

Major Gaps

- **Aberrations:** 0 of 10 (Choker, Grick, Otyugh, Gauth, Mind Flayer, etc.)
- **Lycanthropes:** Template exists but missing 5 core variants (Wererat, Wereboar, etc.)
- **Critical Low-CR Creatures:** Darkmantle, Ghoul, Kobold, standard humanoid warriors
- **Hydras:** Need multi-head system (4 variants)
- **Special Abilities:** Gaze attacks, petrification, energy drain, disease variations

- **Size-Scaled Vermin:** 15 missing (Giant Constrictor Snake, Dire Boar, etc.)

Implementation Philosophy

1. **No modifications to base creatures** — always clone before applying changes
2. **Reuse existing systems** — CreatureTemplateRegistry, CreatureClassEngine, DungeonEncounter-Spawner
3. **Coding style:** No namespaces, 4-space indent, XML doc comments
4. **Alphabetical organization** — Creatures in NPCDatabase_[Letter].cs files
5. **Debug logging** — Use `Debug.Log` for registration tracking

Implementation Phases

Phase 1: Quick Wins (Name Mappings & Simple Stat Blocks)

Goal: Add 30 creatures in 2-3 days
Focus: Simple stat blocks, name mappings, and creatures that require minimal special mechanics
Expected Coverage Improvement: 62.6% → 77.8% (+15.2%)

1.1 Simple Name Mappings (20 creatures, ~4 hours)

These creatures exist but need ID mappings for encounter table compatibility.

Creature	Mapping	File	Effort
Goblin Warrior	<code>goblin</code> → <code>goblin_warrior</code>	NPCDatabase_G.cs	Alias
Orc Warrior	<code>orc_berserker</code> → <code>orc_warrior</code>	NPCDatabase_O.cs	Alias
Hobgoblin Warrior	<code>hobgoblin_sergeant</code> → <code>hobgoblin_warrior</code>	NPCDatabase_H.cs	Alias
Dire Wolf	Existing → <code>dire_wolf_mm</code>	NPCDatabaseCustom.cs	Alias
Brown Bear	Existing → <code>brown_bear_mm</code>	NPCDatabaseCustom.cs	Alias
Human Warrior 1-3	Create variants	NPCDatabase_H.cs	3 entries

Implementation:

```
// In RegisterCreatures_G() – NPCDatabase_G.cs
RegisterSummonAlias("goblin_warrior", "goblin");

// In RegisterCreatures_O() – NPCDatabase_O.cs
RegisterSummonAlias("orc_warrior", "orc_berserker");

// In RegisterCreatures_H() – NPCDatabase_H.cs
RegisterHumanWarrior1();
RegisterHumanWarrior2();
RegisterHumanWarrior3();
RegisterSummonAlias("hobgoblin_warrior", "hobgoblin_sergeant");
```

Dependencies: None

Testing: Verify encounter table lookup, ensure no ID conflicts

Estimated Time: 4 hours

1.2 Simple Animals (5 creatures, ~2 hours)

Basic animal stat blocks with minimal abilities.

Creature	CR	Special Abilities	File
Hyena	1	Trip on bite	NPCDatabase_H.cs
Monitor Lizard	2	Grab	NPCDatabase_M.cs
Giant Constrictor Snake	5	Constrict, improved grab	NPCDatabase_G.cs
Dire Boar	4	Ferocity	NPCDatabase_D.cs
Krenshar	1	Scare (skull illusion)	NPCDatabase_K.cs

Implementation Example (Hyena):

```

/// <summary>
/// Hyena (CR 1) – Medium animal.
/// MM 3.5e p.274. Pack hunter with trip attack.
/// </summary>
private static void RegisterHyena()
{
    Register(new NPCDefinition
    {
        Id = "hyena",
        Name = "Hyena",
        ChallengeRating = "1",
        Level = 2,
        CharacterClass = "Warrior",
        CreatureType = "Animal",
        HitDice = 2,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 14, DEX = 15, CON = 15, WIS = 13, INT = 2, CHA = 6,
        BAB = 1,
        NaturalArmorBonus = 2,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthOneAndHalf,
                Range = 1, IsPrimary = true,
                TripOnHit = true
            }
        },
        BaseSpeed = 10,
        BaseHitDieHP = 13,
        CreatureTags = new List<string> { "Animal", "MM35" },
        Feats = new List<string> { "Alertness" },
        HasScent = true,
        SpecialAbilities = new List<string>
        {
            "Low-light vision",
            "Scent",
            "Trip: On successful bite, can attempt free trip (no AoO)",
            "Skills: Hide +3, Listen +6, Spot +4"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Animal,
        SpriteColor = new Color(0.75f, 0.65f, 0.45f, 1f),
        PanelColor = new Color(0.25f, 0.2f, 0.1f, 0.85f),
        NameColor = new Color(0.95f, 0.88f, 0.7f),
        Description = "Monster Manual hyena. Medium pack hunter with trip attack and
scent tracking."
    });
}

```

Dependencies: Existing NaturalAttackDefinition, trip mechanics

Testing: Verify trip on bite works in combat

Estimated Time: 2 hours

1.3 Basic Humanoids (5 creatures, ~3 hours)

Low-CR humanoid warriors for common encounters.

Creature	CR	Equipment	File
Kobold Warrior	1/4	Spear, sling	NPCDatabase_K.cs
Grimlock	1	Battleaxe, blindsight	NPCDatabase_G.cs
Skum	2	Trident	NPCDatabase_S.cs
Derro	3	Aklys, poison, spell-like	NPCDatabase_D.cs
Human Commoner 1	1/2	Club, basic gear	NPCDatabase_H.cs

Implementation Example (Kobold):

```

/// <summary>
/// Kobold Warrior (CR 1/4) – Small humanoid (reptilian).
/// MM 3.5e p.161. Warrior 1. Light sensitivity, trap-making.
/// </summary>
private static void RegisterKoboldWarrior()
{
    Register(new NPCDefinition
    {
        Id = "kobold_warrior",
        Name = "Kobold Warrior",
        ChallengeRating = "1/4",
        Level = 1,
        CharacterClass = "Warrior",
        CreatureType = "Humanoid",
        HitDice = 1,
        SizeCategory = SizeCategory.Small,
        IsTallCreature = false,
        STR = 9, DEX = 13, CON = 10, WIS = 10, INT = 10, CHA = 8,
        BAB = 0,
        NaturalArmorBonus = 1,
        NaturalAttacks = new List<NaturalAttackDefinition>(),
        BaseSpeed = 6,
        BaseHitDieHP = 4,
        CreatureTags = new List<string> { "Humanoid", "Reptilian", "MM35" },
        Feats = new List<string> { "Alertness" },
        SpecialAbilities = new List<string>
        {
            "Darkvision 60 ft.",
            "Light Sensitivity: -1 penalty to attacks in bright sunlight",
            "Skills: Craft (trapmaking) +2, Hide +6, Listen +2, Move Silently +2,
            Profession (miner) +2, Search +2, Spot +2"
        },
        EquipmentIds = new List<EquipmentSlotPair>
        {
            new EquipmentSlotPair("main_hand", "spear"),
            new EquipmentSlotPair("off_hand", "light_wooden_shield"),
            new EquipmentSlotPair("ranged", "sling")
        },
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.RangedThenMelee,
        AIProfileArchetype = NPCAIProfileArchetype.SkirmisherRanged,
        SpriteColor = new Color(0.6f, 0.45f, 0.3f, 1f),
        PanelColor = new Color(0.2f, 0.15f, 0.1f, 0.85f),
        NameColor = new Color(0.9f, 0.75f, 0.55f),
        Description = "Monster Manual kobold warrior. Small reptilian humanoid with
            light sensitivity and trap expertise."
    });
}

```

Dependencies: Equipment system, light sensitivity mechanics (if implemented)

Testing: Verify equipment loads, AI behavior

Estimated Time: 3 hours

Phase 1 Success Criteria

- [] 30 creatures added to NPCDatabase
- [] All creatures compile without errors

- [] Coverage improves from 62.6% to 77.8%
- [] Encounter tables can resolve all Phase 1 creature IDs
- [] Manual spawn test of 5 random Phase 1 creatures succeeds
- [] Debug.Log shows registration count increase

Phase 1 Effort Summary:

- **Total creatures:** 30
 - **Estimated time:** 2-3 days (16-24 hours)
 - **Files modified:** NPCDatabase_D/G/H/K/M/O/S.cs (7 files)
 - **New systems required:** None
 - **Dependencies:** Existing equipment, AI systems
-

Phase 2: Core Missing Creatures

Goal: Add 25 high-priority creatures in 4-5 days

Focus: Critical dungeon encounter creatures with moderate complexity

Expected Coverage Improvement: 77.8% → 90.4% (+12.6%)

2.1 Aberrations (10 creatures, ~8 hours)

Key aberrations with unique abilities.

Creature	CR	Special Abilities	Complexity	File
Darkmantle	1	Darkness aura, grab	Medium	NPCData-base_D.cs
Choker	2	Reach tentacles, improved grab	Medium	NPCData-base_C.cs
Grick	3	Damage reduction 10/magic	Easy	NPCData-base_G.cs
Otyugh	4	Disease, improved grab, tentacles	Medium	NPCData-base_O.cs
Carrion Crawler	4	Paralysis tentacles (×8)	Medium	NPCData-base_C.cs
Gibbering Mouther	5	Gibbering (confusion), ground manipulation	Hard	NPCData-base_G.cs
Gauth	6	Eye rays (6 types, 1/round)	Hard	NPCData-base_G.cs
Chuul	7	Paralytic tentacles, amphibious	Medium	NPCData-base_C.cs
Destrachan	8	Sonic attacks (3 types)	Medium	NPCData-base_D.cs
Mind Flayer	8	Mind blast, extract brain	Hard	NPCData-base_M.cs

Implementation Example (Darkmantle):


```

/// <summary>
/// Darkmantle (CR 1) – Small magical beast.
/// MM 3.5e p.52. Cave-dwelling ambush predator that creates darkness.
/// </summary>
private static void RegisterDarkmantle()
{
    Register(new NPCDefinition
    {
        Id = "darkmantle",
        Name = "Darkmantle",
        ChallengeRating = "1",
        Level = 1,
        CharacterClass = "Warrior",
        CreatureType = "Magical Beast",
        HitDice = 1,
        SizeCategory = SizeCategory.Small,
        IsTallCreature = false,
        STR = 16, DEX = 15, CON = 13, WIS = 12, INT = 2, CHA = 10,
        BAB = 1,
        NaturalArmorBonus = 5,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Slam", DamageDice = 4, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthOneAndHalf,
                Range = 1, IsPrimary = true
            }
        },
        BaseSpeed = 4,
        BaseHitDieHP = 9,
        CreatureTags = new List<string> { "Magical Beast", "MM35" },
        Feats = new List<string> { "Improved Initiative" },
        HasBlindSense = true,
        BlindSenseRange = 90,
        SpecialAbilities = new List<string>
        {
            "Blindsight 90 ft.",
            "Darkness (Su): Once per day, 20 ft radius for 1 minute (CL 5)",
            "Improved Grab: On slam hit, can start grapple as free action",
            "Skills: Hide +10, Listen +4"
        },
        SpellLikeAbilities = new List<SpellLikeAbilityDefinition>
        {
            new SpellLikeAbilityDefinition
            {
                SpellId = "darkness",
                UsesPerDay = 1,
                CasterLevel = 5
            }
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Ambusher,
        SpriteColor = new Color(0.2f, 0.2f, 0.25f, 1f),
        PanelColor = new Color(0.1f, 0.1f, 0.15f, 0.9f),
        NameColor = new Color(0.6f, 0.6f, 0.7f),
        Description = "Monster Manual darkmantle. Small cave-dweller that cloaks
itself in magical darkness before dropping on prey."
    });
}

```

Implementation Example (Gauth — Eye Rays):

```

/// <summary>
/// Gauth (CR 6) – Medium aberration (beholder-kin).
/// MM 3.5e p.27. Lesser beholder with 6 eye rays and central eye stunning beam.
/// </summary>
private static void RegisterGauth()
{
    Register(new NPCDefinition
    {
        Id = "gauth",
        Name = "Gauth",
        ChallengeRating = "6",
        Level = 6,
        CharacterClass = "Warrior",
        CreatureType = "Aberration",
        HitDice = 6,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 10, DEX = 14, CON = 14, WIS = 14, INT = 13, CHA = 13,
        BAB = 4,
        NaturalArmorBonus = 9,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 4, DamageCount = 2, Count = 1,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 1, IsPrimary = true
            }
        },
        BaseSpeed = 5,
        BaseHitDieHP = 45,
        CreatureTags = new List<string> { "Aberration", "MM35" },
        Feats = new List<string> { "Alertness", "Improved Initiative", "Iron Will" },
        SpecialAbilities = new List<string>
        {
            "All-Around Vision: Cannot be flanked, +4 Spot/Search",
            "Eye Rays (Su): 6 small eyes, each can fire once per round, 100 ft range,
+8 ranged touch",
            " 1. Inflict Moderate Wounds (2d8+6 damage, Will DC 16 half)",
            " 2. Dispel Magic (CL 13, +13 dispel check)",
            " 3. Exhaustion Ray (Fort DC 16 or become exhausted)",
            " 4. Fear Ray (Will DC 16 or flee 1 minute)",
            " 5. Paralyzing Ray (Fort DC 16 or hold person 1d4+1 rounds)",
            " 6. Scorching Ray (4d6 fire damage, ranged touch)",
            "Central Eye (Su): Stunning beam 60 ft cone, 1/round, Fort DC 16 or
stunned 1d4 rounds",
            "Flight 20 ft (good)",
            "Darkvision 60 ft.",
            "Skills: Hide +11, Listen +13, Search +14, Spot +13"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.RangedController,
        AIProfileArchetype = NPCAIProfileArchetype.TacticalCaster,
        SpriteColor = new Color(0.4f, 0.35f, 0.5f, 1f),
        PanelColor = new Color(0.15f, 0.1f, 0.2f, 0.9f),
        NameColor = new Color(0.8f, 0.7f, 0.9f),
        Description = "Monster Manual gauth. Lesser beholder with 6 magical eye rays
and stunning central eye beam."
    });
}

```

Dependencies:

- SpellLikeAbilityDefinition for Darkmantle darkness
- Eye ray attack system for Gauth (may need new AttackType enum)
- Improved grab mechanics

Testing:

- Darkmantle spawns with darkness ability
- Gauth eye rays fire correctly (1/round rotation)
- Mind Flayer mind blast affects multiple targets

Estimated Time: 8 hours

2.2 Undead (5 creatures, ~4 hours)

Critical undead with paralysis, stench, and energy drain.

Creature	CR	Special Abilities	File
Ghoul	1	Paralysis touch, un-dead traits	NPCDatabase_G.cs
Ghast	3	Stench, paralysis, un-dead	NPCDatabase_G.cs
Wight	3	Energy drain (1 level), create spawn	NPCDatabase_W.cs
Vampire Spawn	4	Energy drain (2 levels), dominate	NPCDatabase_V.cs
Mohrg	8	Paralysis tongue, create zombies	NPCDatabase_M.cs

Implementation Example (Ghoul):

```

/// <summary>
/// Ghoul (CR 1) – Medium undead.
/// MM 3.5e p.118. Flesh-eating undead with paralysis touch.
/// </summary>
private static void RegisterGhoul()
{
    Register(new NPCDefinition
    {
        Id = "ghoul",
        Name = "Ghoul",
        ChallengeRating = "1",
        Level = 2,
        CharacterClass = "Warrior",
        CreatureType = "Undead",
        HitDice = 2,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 13, DEX = 15, CON = 0, WIS = 10, INT = 13, CHA = 12,
        BAB = 1,
        NaturalArmorBonus = 2,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 1, IsPrimary = true,
                HasParalysisOnHit = true, ParalysisDurationRounds = 4
            },
            new NaturalAttackDefinition
            {
                Name = "Claw", DamageDice = 3, DamageCount = 1, Count = 2,
                BonusDamageSource = DamageBonusSource.StrengthHalf,
                Range = 1, IsPrimary = false,
                HasParalysisOnHit = true, ParalysisDurationRounds = 4
            }
        },
        BaseSpeed = 6,
        BaseHitDieHP = 13,
        CreatureTags = new List<string> { "Undead", "MM35" },
        Feats = new List<string> { "Multiattack" },
        HasDarkvision = true,
        SpecialAbilities = new List<string>
        {
            "Undead Traits: No Constitution, immunity to mind effects/poison/sleep/paralysis/stunning/disease/death effects",
            "Paralysis (Ex): Bite/claw hit, Fort DC 12 or paralyzed 1d4+1 rounds (elves immune)",
            "+2 turn resistance",
            "Darkvision 60 ft.",
            "Skills: Balance +6, Climb +5, Hide +6, Jump +5, Move Silently +6, Spot +5"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Undead,
        SpriteColor = new Color(0.45f, 0.5f, 0.4f, 1f),
        PanelColor = new Color(0.15f, 0.2f, 0.15f, 0.9f),
        NameColor = new Color(0.75f, 0.85f, 0.7f),
        Description = "Monster Manual ghoul. Flesh-eating undead with paralytic bite and claws."
    }
}

```

```
    });  
}
```

Implementation Example (Ghast — with Stench):

```

/// <summary>
/// Ghast (CR 3) – Medium undead.
/// MM 3.5e p.119. More powerful ghouls with stench ability and immunity to Turn Un-
dead.
/// </summary>
private static void RegisterGhast()
{
    Register(new NPCDefinition
    {
        Id = "ghast",
        Name = "Ghast",
        ChallengeRating = "3",
        Level = 4,
        CharacterClass = "Warrior",
        CreatureType = "Undead",
        HitDice = 4,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 17, DEX = 17, CON = 0, WIS = 10, INT = 13, CHA = 16,
        BAB = 2,
        NaturalArmorBonus = 4,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 1, IsPrimary = true,
                HasParalysisOnHit = true, ParalysisDurationRounds = 6
            },
            new NaturalAttackDefinition
            {
                Name = "Claw", DamageDice = 4, DamageCount = 1, Count = 2,
                BonusDamageSource = DamageBonusSource.StrengthHalf,
                Range = 1, IsPrimary = false,
                HasParalysisOnHit = true, ParalysisDurationRounds = 6
            }
        },
        BaseSpeed = 6,
        BaseHitDieHP = 26,
        CreatureTags = new List<string> { "Undead", "MM35" },
        Feats = new List<string> { "Multiattack", "Toughness" },
        HasDarkvision = true,
        SpecialAbilities = new List<string>
        {
            "Undead Traits",
            "Stench (Ex): 10 ft aura. Fort DC 15 or sickened for 1d6+4 minutes",
            "Paralysis (Ex): Bite/claw hit, Fort DC 15 or paralyzed 1d4+4 rounds (no
elf immunity)",
            "+4 turn resistance",
            "Darkvision 60 ft.",
            "Skills: Balance +7, Climb +7, Hide +7, Jump +7, Move Silently +7, Spot
+7"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Undead,
        SpriteColor = new Color(0.35f, 0.4f, 0.35f, 1f),
        PanelColor = new Color(0.12f, 0.15f, 0.12f, 0.92f),
        NameColor = new Color(0.65f, 0.75f, 0.65f),
        Description =

```

```
"Monster Manual ghast. Powerful ghoul with nauseating stench and paralytic attacks."
    });
}
```

Dependencies:

- Paralysis system (HasParalysisOnHit flag)
- Energy drain mechanics (for Wight/Vampire Spawn)
- Stench aura system (new)

Testing: Ghoul paralysis works on hit, Ghast stench affects nearby creatures

Estimated Time: 4 hours

2.3 Magical Beasts (6 creatures, ~5 hours)

Iconic dungeon creatures.

Creature	CR	Special Abilities	File
Owlbear	4	Improved grab	NPCDatabase_O.cs
Ankheg	3	Spit acid, improved grab	NPCDatabase_A.cs
Basilisk	5	Petrifying gaze	NPCDatabase_B.cs
Displacer Beast	4	Displacement (50% miss)	NPCDatabase_D.cs
Manticore	5	Tail spike volley	NPCDatabase_M.cs
Phase Spider	5	Ethereal jaunt	NPCDatabase_P.cs

Implementation Example (Owlbear):


```

/// <summary>
/// Owlbear (CR 4) – Large magical beast.
/// MM 3.5e p.206. Bear-owl hybrid, aggressive and territorial.
/// </summary>
private static void RegisterOwlbear()
{
    Register(new NPCDefinition
    {
        Id = "owlbear",
        Name = "Owlbear",
        ChallengeRating = "4",
        Level = 5,
        CharacterClass = "Warrior",
        CreatureType = "Magical Beast",
        HitDice = 5,
        SizeCategory = SizeCategory.Large,
        IsTallCreature = true,
        STR = 21, DEX = 12, CON = 17, WIS = 12, INT = 2, CHA = 10,
        BAB = 5,
        NaturalArmorBonus = 5,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Claw", DamageDice = 6, DamageCount = 1, Count = 2,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 2, IsPrimary = true
            },
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 8, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthHalf,
                Range = 2, IsPrimary = false
            }
        },
        BaseSpeed = 6,
        BaseHitDieHP = 42,
        CreatureTags = new List<string> { "Magical Beast", "MM35" },
        Feats = new List<string> { "Alertness", "Track" },
        HasScent = true,
        SpecialAbilities = new List<string>
        {
            "Low-light vision",
            "Scent",
            "Improved Grab (Ex): On claw hit, can start grapple as free action. If
both claws hit, gets +4 to grapple and can rake on next round.",
            "Skills: Listen +8, Spot +8"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Brute,
        SpriteColor = new Color(0.55f, 0.45f, 0.35f, 1f),
        PanelColor = new Color(0.2f, 0.15f, 0.1f, 0.88f),
        NameColor = new Color(0.85f, 0.75f, 0.65f),
        Description = "Monster Manual owlbear. Large territorial predator with
ferocious grappling attacks."
    });
}

```

Implementation Example (Basilisk — Gaze Attack):

```

/// <summary>
/// Basilisk (CR 5) – Medium magical beast.
/// MM 3.5e p.23. Reptilian creature with petrifying gaze.
/// </summary>
private static void RegisterBasilisk()
{
    Register(new NPCDefinition
    {
        Id = "basilisk",
        Name = "Basilisk",
        ChallengeRating = "5",
        Level = 6,
        CharacterClass = "Warrior",
        CreatureType = "Magical Beast",
        HitDice = 6,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 15, DEX = 8, CON = 15, WIS = 12, INT = 2, CHA = 11,
        BAB = 6,
        NaturalArmorBonus = 7,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 8, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthOneAndHalf,
                Range = 1, IsPrimary = true
            }
        },
        BaseSpeed = 4,
        BaseHitDieHP = 45,
        CreatureTags = new List<string> { "Magical Beast", "MM35" },
        Feats = new List<string> { "Alertness", "Blind-Fight", "Great Fortitude" },
        SpecialAbilities = new List<string>
        {
            "Petrifying Gaze (Su): 30 ft range, Fort DC 13 or turn to stone perman-
ently",
            "Gaze can be avoided by averting eyes (-4 to attack basilisk) or closing
eyes (50% miss chance, lose Dex bonus to AC)",
            "Basilisk's gaze also affects creatures with darkvision in darkness",
            "Darkvision 60 ft.",
            "Low-light vision",
            "Skills: Listen +7, Spot +7"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Bruiser,
        SpriteColor = new Color(0.5f, 0.55f, 0.3f, 1f),
        PanelColor = new Color(0.18f, 0.2f, 0.1f, 0.9f),
        NameColor = new Color(0.8f, 0.85f, 0.6f),
        Description = "Monster Manual basilisk. Eight-legged reptile whose gaze turns
victims to stone."
    });
}

```

Dependencies:

- Improved grab (existing)
- Gaze attack system (new — needs GazeAttackDefinition)
- Displacement visual effect (for Displacer Beast)

Testing: Owlbear grapple works, Basilisk gaze triggers save

Estimated Time: 5 hours

2.4 Oozes & Plants (4 creatures, ~3 hours)

Dungeon hazards with unique damage types.

Creature	CR	Special Abilities	File
Grey Ooze	4	Acid, corrosion	NPCDatabase_G.cs
Black Pudding	7	Acid, split, corrosion	NPCDatabase_B.cs
Violet Fungus	3	Rotting touch tentacles	NPCDatabase_V.cs
Phantom Fungus	3	Invisibility	NPCDatabase_P.cs

Implementation Example (Grey Ooze):

```

/// <summary>
/// Grey Ooze (CR 4) – Medium ooze.
/// MM 3.5e p.202. Acidic ooze that corrodes metal.
/// </summary>
private static void RegisterGreyOoze()
{
    Register(new NPCDefinition
    {
        Id = "grey_ooze",
        Name = "Grey Ooze",
        ChallengeRating = "4",
        Level = 3,
        CharacterClass = "Warrior",
        CreatureType = "Ooze",
        HitDice = 3,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 12, DEX = 1, CON = 21, WIS = 1, INT = 0, CHA = 1,
        BAB = 2,
        NaturalArmorBonus = 0,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Slam", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthOneAndHalf,
                DamageType = DamageType.Acid,
                Range = 1, IsPrimary = true,
                AdditionalEffect = "1d6 acid damage"
            }
        },
        BaseSpeed = 2,
        BaseHitDieHP = 34,
        CreatureTags = new List<string> { "Ooze", "MM35" },
        Feats = new List<string>(),
        HasBlindSight = true,
        BlindSightRange = 60,
        SpecialAbilities = new List<string>
        {
            "Ooze Traits: Immune to mind effects, poison, sleep, paralysis, polymorph, stunning,
            critical hits, flanking",
            "Acid (Ex): 1d6 acid damage on slam hit plus 1d6 per round to armor/cloth-
            ing",
            "Constrict (Ex): Automatic slam damage on successful grapple",
            "Improved Grab (Ex): On slam hit, can start grapple",
            "Transparent (Ex): DC 15 Spot check to notice when motionless",
            "Blindsight 60 ft.",
            "Immune to cold and fire"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Mindless,
        SpriteColor = new Color(0.5f, 0.5f, 0.52f, 1f),
        PanelColor = new Color(0.25f, 0.25f, 0.27f, 0.85f),
        NameColor = new Color(0.7f, 0.7f, 0.75f),
        Description = "Monster Manual grey ooze. Acidic ooze that corrodes metal and
        is nearly transparent."
    });
}

```

Dependencies: Acid damage type, transparent mechanics

Testing: Acid damage applies to equipment

Estimated Time: 3 hours

Phase 2 Success Criteria

- [] 25 creatures added (total: 55 of 74)
- [] Coverage improves to 90.4%
- [] All aberrations compile and spawn correctly
- [] Ghoul paralysis and Ghast stench work in combat
- [] Basilisk gaze triggers petrification save
- [] Eye rays (Gauth) fire 1/round
- [] Ooze acid damage applies correctly
- [] Encounter tables resolve all Phase 2 creature IDs

Phase 2 Effort Summary:

- **Total creatures:** 25
- **Estimated time:** 4-5 days (32-40 hours)
- **Files modified:** NPCDatabase_A/B/C/D/G/K/M/O/P/V/W.cs (11 files)
- **New systems required:**
 - Gaze attack system (GazeAttackDefinition)
 - Stench aura system
 - Eye ray rotation system
 - Energy drain mechanics (if not existing)
- **Dependencies:** Existing paralysis, improved grab, blindsight

Phase 3: Template Systems

Goal: Implement 3 template systems and 13 variants in 5-6 days

Focus: Lycanthrope, Ghost, and multi-head systems

Expected Coverage Improvement: 90.4% → 96.9% (+6.5%)

3.1 Lycanthrope Template Completion (5 creatures, ~12 hours)

The lycanthrope template framework exists. Need to add missing variants.

Existing: LycanthropeTemplate.cs, LycanthropeFactory with Werewolf/Werewolf Lord/Wererat/Wereboar/Weretiger/Werebear (already done per context)

Missing Variants:

Creature	CR	Base Form	Animal Form	Status	
-----	----	-----	-----	-----	
Wererat	2	Human Rogue 1	Dire Rat	☒ Exists	
Werewolf	3	Human Warrior 1	Wolf	☒ Exists	
Wereboar	4	Human Barbarian 1	Boar	☒ Exists	
Werebear	5	Human Commoner 1	Brown Bear	☒ Exists	
Weretiger	5	Human Noble 4	Tiger	☒ Exists	

NOTE: Based on NPCDatabase_Lycanthropes.cs review, all 5 core lycanthropes are already implemented via LycanthropeFactory! This phase focuses on:

1. Verifying template consistency
2. Adding lycanthrope encounter presets (already exist)
3. Creating “animal form” and “hybrid form” variants if needed

Revised Phase 3.1: Add Dire Wereboar and Afflicted variants (already done), focus on expanding template to other base creatures.

New Goal: Add 3 custom lycanthrope variants using existing template:

- Werewolf (Elf Ranger 3 base)
- Wererat (Halfling Rogue 2 base)
- Werebear (Dwarf Fighter 4 base)

Implementation Example (Custom Werewolf):

```
// In LycanthropeFactory.cs, add new method:

/// <summary>
/// Elf Ranger Werewolf (CR 5) – Natural lycanthrope.
/// Elf ranger 3 + wolf hybrid form.
/// </summary>
public static NPCDefinition ElfRangerWerewolf()
{
    // Clone base elf ranger 3 (would need to create this first)
    NPCDefinition baseElf = NPCDatabase.Get("elf_ranger_3").Clone();
    NPCDefinition wolfBase = NPCDatabase.Get("wolf").Clone();

    return LycanthropeTemplate.Apply(
        baseHumanoid: baseElf,
        animalBase: wolfBase,
        lycanthropeId: "elf_ranger_werewolf",
        lycanthropeName: "Elf Ranger Werewolf",
        isAfflicted: false,
        challengeRating: "5"
    );
}

// Then register in NPCDatabase_Lycanthropes.cs:
Register(LycanthropeFactory.ElfRangerWerewolf());
```

Dependencies:

- LycanthropeTemplate.Apply() (exists)
- Base creatures (elf_ranger_3, halfling_rogue_2, dwarf_fighter_4 — need to create)

Testing:

- Custom lycanthrope spawns with correct stats
- DR 10/silver applies
- Hybrid form stats merge correctly

Estimated Time: 12 hours (includes creating 3 base NPC classes)

3.2 Hydra Multi-Head System (4 creatures, ~16 hours)

Hydras need a parametric system for variable head count affecting attacks and HP.

Creature	CR	Heads	Breath	File
5-headed Hydra	4	5	No	NPCData-base_H.cs
6-headed Hydra	5	6	No	NPCData-base_H.cs
7-headed Pyro-hydra	8	7	Fire	NPCData-base_H.cs
8-headed Hydra	7	8	No	NPCData-base_H.cs

Design: Create HydraFactory with parametric head generation.

New File: Assets/Scripts/Character/HydraFactory.cs

```

using System.Collections.Generic;
using UnityEngine;

/// <summary>
/// Factory for generating D&D 3.5e hydras with variable head counts.
/// MM 3.5e p.155-157.
///
/// Design:
/// - Base stats scale with head count
/// - Each head = 1 bite attack
/// - HD = 5 + head count
/// - CR scales: 5-head (CR 4), 6-head (CR 5), 7-head (CR 6), 8-head (CR 7)
/// - Pyrohydra variant has fire immunity and breath weapon
///
/// Fast Healing scales with heads:
/// - Standard: 10 + (5 × heads)
/// - Pyrohydra: 15 + (5 × heads)
/// </summary>
public static class HydraFactory
{
    /// <summary>
    /// Generate a standard hydra with specified head count.
    /// </summary>
    public static NPCDefinition CreateHydra(int headCount)
    {
        if (headCount < 5 || headCount > 12)
        {
            Debug.LogWarning($"[HydraFactory] Invalid head count {headCount}. Clamping to 5-12.");
            headCount = Mathf.Clamp(headCount, 5, 12);
        }

        int hd = 5 + headCount;
        int cr = headCount - 1; // 5-head = CR 4, 6-head = CR 5, etc.

        // STR scales slightly with size
        int str = 17 + (headCount - 5) / 2;

        // Natural attacks: 1 bite per head
        List<NaturalAttackDefinition> bites = new List<NaturalAttackDefinition>();
        for (int i = 0; i < headCount; i++)
        {
            bites.Add(new NaturalAttackDefinition
            {
                Name = "Bite",
                DamageDice = 10,
                DamageCount = 1,
                Count = 1,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 2,
                IsPrimary = true
            });
        }

        return new NPCDefinition
        {
            Id = $"hydra_{headCount}head",
            Name = $"{headCount}-Headed Hydra",
            ChallengeRating = cr.ToString(),
            Level = hd,
            CharacterClass = "Warrior",
            CreatureType = "Magical Beast",
        }
    }
}

```



```

        HitDice = hd,
        SizeCategory = SizeCategory.Huge,
        IsTallCreature = true,
        STR = str, DEX = 12, CON = 20, WIS = 10, INT = 2, CHA = 9,
        BAB = hd,
        NaturalArmorBonus = 5,
        NaturalAttacks = bites,
        BaseSpeed = 4,
        BaseHitDieHP = (hd * 7) + (hd * 5), // d10 average = 7, +5 Con bonus
        CreatureTags = new List<string> { "Magical Beast", "MM35", "Hydra" },
        Feats = GenerateHydraFeats(headCount),
        SpecialAbilities = new List<string>
        {
            $"Combat Reflexes: Can make {headCount} AoOs per round",

            $"Fast Healing {10 + (headCount * 5)}: Regrows heads unless cauterized with fire/acid",
            "Low-light vision",
            "Scent",
            $"Skills: Listen +{hd + 2}, Spot +{hd + 2}, Swim +{hd + 8}"
        },
        HasScent = true,
        HasFastHealing = true,
        FastHealingRate = 10 + (headCount * 5),
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Brute,
        SpriteColor = new Color(0.4f, 0.6f, 0.3f, 1f),
        PanelColor = new Color(0.15f, 0.25f, 0.1f, 0.88f),
        NameColor = new Color(0.7f, 0.9f, 0.6f),
        Description =
            $"Monster Manual {headCount}-headed hydra. Huge multi-headed serpent with regenerating heads and relentless attacks."
    };
}

/// <summary>
/// Generate a pyrohydra (fire-breathing variant).
/// </summary>
public static NPCDefinition CreatePyrohydra(int headCount)
{
    NPCDefinition hydra = CreateHydra(headCount);

    hydra.Id = $"pyrohydra_{headCount}head";
    hydra.Name = $"{headCount}-Headed Pyrohydra";
    hydra.ChallengeRating = (headCount + 1).ToString(); // +1 CR for fire breath

    // Pyrohydra has better fast healing
    hydra.FastHealingRate = 15 + (headCount * 5);

    // Add fire immunity and breath weapon
    hydra.SpecialAbilities.Add("Fire Immunity");
    hydra.SpecialAbilities.Add($"Breath Weapon (Su): Each head can breathe 10 ft cone of fire (3d6 damage, Reflex DC {12 + headCount} half) 1/day");
    hydra.SpecialAbilities[0] = $"Combat Reflexes: Can make {headCount} AoOs per round";
    hydra.SpecialAbilities[1] = $"Fast Healing {15 + (headCount * 5)}: Regrows heads (not cauterized by fire)";

    hydra.CreatureTags.Add("Fire");
    hydra.SpriteColor = new Color(0.8f, 0.4f, 0.2f, 1f);
    hydra.PanelColor = new Color(0.3f, 0.15f, 0.1f, 0.88f);

```

```

        hydra.NameColor = new Color(1f, 0.7f, 0.4f);
        hydra.Description = $"Monster Manual {headCount}-headed pyrohydra. Fire-breathing hydra immune to flames.";

        return hydra;
    }

    private static List<string> GenerateHydraFeats(int headCount)
    {
        List<string> feats = new List<string>
        {
            "Combat Reflexes",
            "Iron Will",
            "Toughness"
        };

        // Additional feats based on HD
        int hd = 5 + headCount;
        if (hd >= 9) feats.Add("Weapon Focus (bite)");
        if (hd >= 12) feats.Add("Improved Initiative");

        return feats;
    }

    // — Standard MM hydra presets —

    public static NPCDefinition Hydra5Head() => CreateHydra(5);
    public static NPCDefinition Hydra6Head() => CreateHydra(6);
    public static NPCDefinition Hydra7Head() => CreateHydra(7);
    public static NPCDefinition Hydra8Head() => CreateHydra(8);

    public static NPCDefinition Pyrohydra7Head() => CreatePyrohydra(7);
}

```

Register in NPCDatabase_H.cs:

```

private static void RegisterCreatures_H()
{
    // ... existing registrations ...

    // Hydras
    Register(HydraFactory.Hydra5Head());
    Register(HydraFactory.Hydra6Head());
    Register(HydraFactory.Hydra7Head());
    Register(HydraFactory.Hydra8Head());
    Register(HydraFactory.Pyrohydra7Head());

    Debug.Log("[NPCDatabase] Registered 5 hydra variants.");
}

```

Dependencies:

- Fast healing system (HasFastHealing, FastHealingRate properties)
- Multiple identical attacks (NaturalAttackDefinition list)
- Combat Reflexes feat (for AoO limit)

Testing:

- 5-head hydra has 5 bite attacks
- Fast healing applies each round

- Pyrohydra immune to fire
- CR scales correctly

Estimated Time: 16 hours (includes HydraFactory creation, testing, integration)

3.3 Ghost Template (4 creatures, ~12 hours)

Ghost template applies incorporeal undead properties to any creature.

Design: Create GhostTemplate that takes base creature and converts to incorporeal undead.

Target Variants:

- Ghost Human Fighter 3 (CR 5)
- Ghost Elf Wizard 5 (CR 7)
- Ghost Dwarf Cleric 7 (CR 9)
- Ghost Ogre (CR 6)

New File: `Assets/Scripts/Character/Templates/GhostTemplate.cs`

```

using System;
using System.Collections.Generic;
using UnityEngine;

/// <summary>
/// Ghost creature template for D&D 3.5e (MM p.117).
/// Converts any creature to an incorporeal undead with horrific appearance,
/// manifestation, and corrupting touch.
///
/// Template Changes:
/// - Type → Undead (retains all subtypes)
/// - Incorporeal: 50% miss chance, ignores AC/natural armor, flies
/// - +4 Dex, +3 Cha
/// - Turn resistance +4
/// - Horrific Appearance (Su): Fort save or age 1d4×10 years + frightened
/// - Corrupting Touch (Su): Touch attack ignores armor, Fortitude save or 1d4 Cha
  damage + 1d6 Con damage
/// - Manifestation: Can become visible/invisible at will
/// - Rejuvenation: Reforms 2d4 days after destruction unless laid to rest
/// - CR: Base +2
/// </summary>
public class GhostTemplate : ICreatureTemplate
{
    public string TemplateId => "ghost";
    public int ApplicationOrder => 10; // Applied early (undead conversion)

    public void ApplyToDefinition(NPCDefinition definition)
    {
        if (definition == null)
            return;

        AddTemplateId(definition, "ghost");

        // — Type change —
        definition.CreatureType = "Undead";
        if (!definition.CreatureTags.Contains("Incorporeal"))
            definition.CreatureTags.Add("Incorporeal");

        // — Ability score adjustments —
        definition.DEX += 4;
        definition.CHA += 3;
        definition.CON = 0; // Undead have no Constitution

        // — Incorporeal traits —
        definition.IsIncorporeal = true;
        definition.NaturalArmorBonus = 0; // Incorporeal ignores natural armor

        // — Turn resistance —
        definition.TurnResistance = (definition.TurnResistance ?? 0) + 4;

        // — Special abilities —
        int cr = CalculateCR(definition);
        int saveDC = 10 + (cr / 2) + GetChaModifier(definition.CHA);

        if (definition.SpecialAbilities == null)
            definition.SpecialAbilities = new List<string>();

        definition.SpecialAbilities.Insert(0, "Undead Traits: Immune to mind effects,
        poison, sleep, paralysis, stunning, disease, death effects, critical hits, nonlethal
        damage, ability drain, energy drain, fatigue, exhaustion, any effect requiring
        Fortitude save (unless affects objects)");
        definition.SpecialAbilities.Insert(1, "Incorporeal: 50% miss chance from non-

```

```

magical attacks. Can pass through solid objects. Immune to non-magical weapons.");
    definition.SpecialAbilities.Insert(2, $"Horrific Appearance (Su): Living
creatures within 60 ft that see ghost must make Fort DC {saveDC} or age 1d4×10 years
and become frightened for 1d4+1 rounds");
    definition.SpecialAbilities.Insert(3, $"Corrupting Touch (Su): Melee touch at-
tack, Fort DC {saveDC} or 1d4 Cha damage + 1d6 temporary Con damage (undead creatures
take 2d4 damage instead, no save)");
    definition.SpecialAbilities.Insert(4,
"Manifestation (Su): Can become ethereal or manifest as visible incorporeal form");
    definition.SpecialAbilities.Insert(5, "Rejuvenation (Su): Destroyed ghost re-
turns 2d4 days later unless remains properly laid to rest");
    definition.SpecialAbilities.Insert(6, "+4 turn resistance");

    // — Natural attacks → touch attacks —
    if (definition.NaturalAttacks != null)
    {
        foreach (var attack in definition.NaturalAttacks)
        {
            attack.DamageType = DamageType.None; // Touch ignores physical damage
            attack.AdditionalEffect = $"Corrupting touch (Fort DC {saveDC})";
        }
    }

    // — Feats: Add Alertness if not present —
    if (definition.Feats == null)
        definition.Feats = new List<string>();
    if (!definition.Feats.Contains("Alertness"))
        definition.Feats.Add("Alertness");

    // — CR adjustment —
    int baseCR = ParseCR(definition.ChallengeRating);
    definition.ChallengeRating = (baseCR + 2).ToString();

    // — Visual —
    definition.SpriteColor = new Color(0.6f, 0.7f, 0.8f, 0.5f); // Translucent
blue-white
    definition.PanelColor = new Color(0.15f, 0.2f, 0.3f, 0.75f);
    definition.NameColor = new Color(0.8f, 0.9f, 1f);

    // — AI —
    definition.AIBehavior = NPCAIBehavior.TacticalMelee;
    definition.AIProfileArchetype = NPCAIProfileArchetype.Undead;

    definition.Description = $"{definition.Name} (Ghost). {defini-
tion.Description}";

    Debug.Log($"[GhostTemplate] Applied to {definition.Name}. CR: {defini-
tion.ChallengeRating}");
}

private static void AddTemplateId(NPCDefinition definition, string templateId)
{
    if (definition.AppliedTemplateIds == null)
        definition.AppliedTemplateIds = new List<string>();
    if (!definition.AppliedTemplateIds.Contains(templateId))
        definition.AppliedTemplateIds.Add(templateId);
}

private static int CalculateCR(NPCDefinition definition)
{
    return ParseCR(definition.ChallengeRating);
}

```

```

private static int ParseCR(string cr)
{
    if (string.IsNullOrEmpty(cr)) return 1;
    if (cr.Contains("/")) return 0; // Fractional CR
    if (int.TryParse(cr, out int result)) return result;
    return 1;
}

private static int GetChaModifier(int cha)
{
    return (cha - 10) / 2;
}
}

```

Register in CreatureTemplateRegistry.cs:

```

private static readonly Dictionary<string, ICreatureTemplate> _templates = new Dic-
tionary<string, ICreatureTemplate>(StringComparer.OrdinalIgnoreCase)
{
    // ... existing templates ...
    { "ghost", new GhostTemplate() },
};

```

Create Ghost Variants in NPCDatabase_G.cs:

```

/// <summary>
/// Ghost template creature registrations.
/// </summary>
private static void RegisterGhostCreatures()
{
    // Ghost Human Fighter 3
    NPCDefinition ghostFighter = NPCDatabase.Get("human_fighter_3").Clone();
    ghostFighter.AppliedTemplateIds = new List<string> { "ghost" };
    ghostFighter.Id = "ghost_fighter_3";
    ghostFighter.Name = "Ghost Fighter";
    Register(CreatureTemplateRegistry.ApplyTemplatesClone(ghostFighter));

    // Ghost Elf Wizard 5
    NPCDefinition ghostWizard = NPCDatabase.Get("elf_wizard_5").Clone();
    ghostWizard.AppliedTemplateIds = new List<string> { "ghost" };
    ghostWizard.Id = "ghost_wizard_5";
    ghostWizard.Name = "Ghost Wizard";
    Register(CreatureTemplateRegistry.ApplyTemplatesClone(ghostWizard));

    // ... etc.

    Debug.Log("[NPCDatabase] Registered 4 ghost template variants.");
}

```

Dependencies:

- IsIncorporeal property
- Incorporeal combat mechanics
- Touch attack system
- Ability score damage (Cha, Con)
- Rejuvenation mechanics

Testing:

- Ghost spawns as incorporeal
- 50% miss chance applies
- Corrupting touch triggers save
- Horrific appearance affects nearby creatures

Estimated Time: 12 hours

Phase 3 Success Criteria

- [] HydraFactory creates 4 hydra variants with correct head count
- [] Lycanthrope template supports 3 new custom variants
- [] GhostTemplate applies incorporeal undead traits correctly
- [] Coverage improves to 96.9% (total: 68 of 74)
- [] Hydra multi-bite attacks display correctly
- [] Ghost horrific appearance triggers within 60 ft
- [] All templates compile and integrate with DungeonEncounterSpawner

Phase 3 Effort Summary:

- **Total creatures:** 13 (3 lycanthropes + 5 hydras + 4 ghosts)
 - **Estimated time:** 5-6 days (40-48 hours)
 - **Files created:** HydraFactory.cs, GhostTemplate.cs
 - **Files modified:** NPCDatabase_H.cs, NPCDatabase_G.cs, NPCDatabase_Lycanthropes.cs, CreatureTemplateRegistry.cs
 - **New systems required:**
 - Multi-head attack generation
 - Ghost incorporeal mechanics
 - Ability score damage tracking
 - **Dependencies:** Existing template framework, fast healing system
-

Phase 4: Advanced Features (Size Scaling & Class-Leveled NPCs)

Goal: Add final 6 creatures with advanced mechanics in 2-3 days

Focus: Size-scaled vermin, class-leveled humanoid NPCs

Expected Coverage Improvement: 96.9% → 100% (+3.1%)

4.1 Size-Scaled Vermin (3 creatures, ~4 hours)

Using existing vermin scaling pattern from centipedes/scorpions/spiders.

Creature	Base CR	Sizes Needed	File
Locust Swarm	3	Swarm (10 ft square)	NPCDatabase_L.cs
Centipede Swarm	4	Swarm (10 ft square)	NPCDatabase_C.cs
Hellwasp Swarm	12	Swarm (10 ft square)	NPCDatabase_H.cs

Implementation Example (Locust Swarm):

```

/// <summary>
/// Locust Swarm (CR 3) – Diminutive vermin swarm.
/// MM 3.5e p.237. Dense cloud of biting insects.
/// </summary>
private static void RegisterLocustSwarm()
{
    Register(new NPCDefinition
    {
        Id = "locust_swarm",
        Name = "Locust Swarm",
        ChallengeRating = "3",
        Level = 6,
        CharacterClass = "Warrior",
        CreatureType = "Vermin",
        HitDice = 6,
        SizeCategory = SizeCategory.Diminutive, // Swarm
        IsTallCreature = false,
        STR = 1, DEX = 17, CON = 10, WIS = 10, INT = 0, CHA = 2,
        BAB = 4,
        NaturalArmorBonus = 0,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Swarm", DamageDice = 6, DamageCount = 2, Count = 1,
                BonusDamageSource = DamageBonusSource.None,
                Range = 0, IsPrimary = true,
                IsSwarmAttack = true
            }
        },
        BaseSpeed = 4,
        BaseHitDieHP = 27,
        CreatureTags = new List<string> { "Vermin", "Swarm", "MM35" },
        Feats = new List<string>(),
        HasDarkvision = true,
        SpecialAbilities = new List<string>
        {
            "Swarm Traits: Half damage from slashing/piercing, immune to weapon damage, grappling, tripping. Takes 1.5× damage from area effects.",
            "Distraction (Ex): Living creature in swarm must make Fort DC 12 or nauseated for 1 round",
            "Swarm Attack: 2d6 damage to all creatures in swarm's space",
            "Immune to mind effects (mindless)",
            "Darkvision 60 ft."
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIPProfileArchetype.Mindless,
        SpriteColor = new Color(0.65f, 0.55f, 0.3f, 0.8f),
        PanelColor = new Color(0.25f, 0.2f, 0.1f, 0.75f),
        NameColor = new Color(0.9f, 0.8f, 0.5f),
        Description = "Monster Manual locust swarm. Cloud of biting insects that engulfs and nauseates victims."
    });
}

```


Dependencies:

- Swarm mechanics (IsSwarmAttack, swarm traits)
- Distraction effect (nausea condition)

Testing: Swarm damages all creatures in space, distraction triggers save

Estimated Time: 4 hours

4.2 Class-Leveled Humanoid NPCs (8 creatures, ~8 hours)

Standard racial warriors for encounter variety.

Creature	CR	Class	Equipment	File
Elf Warrior 1	1/2	Warrior	Longsword, longbow	NPCData-base_E.cs
Dwarf Warrior 1	1/2	Warrior	Waraxe, chain-mail	NPCData-base_D.cs
Halfling Warrior 1	1/2	Warrior	Short sword, sling	NPCData-base_H.cs
Drow Elf Warrior	1	Warrior	Rapier, hand crossbow, SR	NPCData-base_D.cs
Duergar Warrior	1	Warrior	Warhammer, en-large	NPCData-base_D.cs
Svirfneblin Warrior	1	Warrior	Pick, spell-like, SR	NPCData-base_S.cs
Hill Giant	7	Warrior	Rock throwing, greatclub	NPCData-base_H.cs
Stone Giant	8	Warrior	Rock throwing, greatclub	NPCData-base_S.cs

Implementation Example (Drow Elf Warrior):

```

/// <summary>
/// Drow Elf Warrior (CR 1) – Medium humanoid (elf).
/// MM 3.5e p.102. Warrior 1. Spell resistance, spell-like abilities, light blindness.
/// </summary>
private static void RegisterDrowWarrior()
{
    Register(new NPCDefinition
    {
        Id = "drow_warrior",
        Name = "Drow Warrior",
        ChallengeRating = "1",
        Level = 1,
        CharacterClass = "Warrior",
        CreatureType = "Humanoid",
        HitDice = 1,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 11, DEX = 15, CON = 10, WIS = 11, INT = 13, CHA = 10,
        BAB = 1,
        NaturalArmorBonus = 0,
        NaturalAttacks = new List<NaturalAttackDefinition>(),
        BaseSpeed = 6,
        BaseHitDieHP = 8,
        CreatureTags = new List<string> { "Humanoid", "Elf", "MM35" },
        Feats = new List<string> { "Weapon Finesse" },
        SpellResistance = 11 + 1, // SR 11 + class level
        HasDarkvision = true,
        SpecialAbilities = new List<string>
        {
            "Darkvision 120 ft.",

            "Light Blindness: Abrupt exposure to bright light blinds drow for 1 round; -1
            circumstance penalty in bright light",
            "Spell-Like Abilities (Sp): 1/day–dancing lights, darkness, faerie fire
            (CL 1)",
            "Spell Resistance 12",
            "Immune to magic sleep",
            "+2 racial bonus on saves vs enchantment",
            "+2 racial bonus on Listen, Search, Spot checks",
            "Skills: Listen +2, Search +3, Spot +2"
        },
        SpellLikeAbilities = new List<SpellLikeAbilityDefinition>
        {
            new SpellLikeAbilityDefinition { SpellId = "dancing_lights", UsesPerDay =
            1, CasterLevel = 1 },
            new SpellLikeAbilityDefinition { SpellId = "darkness", UsesPerDay = 1, Cas
            terLevel = 1 },
            new SpellLikeAbilityDefinition { SpellId = "faerie_fire", UsesPerDay = 1,
            CasterLevel = 1 }
        },
        EquipmentIds = new List<EquipmentSlotPair>
        {
            new EquipmentSlotPair("main_hand", "rapier"),
            new EquipmentSlotPair("off_hand", "buckler"),
            new EquipmentSlotPair("ranged", "hand_crossbow"),
            new EquipmentSlotPair("armor", "chainmail")
        },
        BackpackItemIds = new List<string> { "drow_poison" },
        AIBehavior = NPCAIBehavior.RangedThenMelee,
        AIProfileArchetype = NPCAIPProfileArchetype.TacticalWarrior,
        SpriteColor = new Color(0.3f, 0.3f, 0.35f, 1f),
        PanelColor = new Color(0.1f, 0.1f, 0.15f, 0.9f),
    }
}

```

```
NameColor = new Color(0.85f, 0.85f, 0.95f),  
Description =  
"Monster Manual drow warrior. Dark elf with spell resistance and light blindness."  
});  
}
```

Implementation Example (Hill Giant):

```

/// <summary>
/// Hill Giant (CR 7) – Large giant.
/// MM 3.5e p.123. Huge humanoid with rock throwing and powerful club attacks.
/// </summary>
private static void RegisterHillGiant()
{
    Register(new NPCDefinition
    {
        Id = "hill_giant",
        Name = "Hill Giant",
        ChallengeRating = "7",
        Level = 12,
        CharacterClass = "Warrior",
        CreatureType = "Giant",
        HitDice = 12,
        SizeCategory = SizeCategory.Large,
        IsTallCreature = true,
        STR = 25, DEX = 8, CON = 19, WIS = 10, INT = 6, CHA = 7,
        BAB = 9,
        NaturalArmorBonus = 6,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Slam", DamageDice = 4, DamageCount = 1, Count = 2,
                BonusDamageSource = DamageBonusSource.Strength,
                Range = 2, IsPrimary = true
            }
        },
        BaseSpeed = 8,
        BaseHitDieHP = 102,
        CreatureTags = new List<string> { "Giant", "MM35" },
        Feats = new List<string> { "Cleave", "Improved Bull Rush", "Power Attack", "We
apon Focus (greatclub)" },
        HasLowLightVision = true,
        SpecialAbilities = new List<string>
        {
            "Rock Throwing (Ex): 100 ft range, 2d6+10 damage",
            "Rock Catching (Ex): Can catch Small, Medium, or Large rocks (or similar
projectiles)",
            "Low-light vision",
            "Skills: Climb +10, Jump +5, Listen +2, Spot +6"
        },
        RangedAttacks = new List<RangedAttackDefinition>
        {
            new RangedAttackDefinition
            {
                Name = "Rock", Range = 100, DamageDice = 6, DamageCount = 2,
                BonusDamage = 10, AmmoCount = 999 // Unlimited rocks
            }
        },
        EquipmentIds = new List<EquipmentSlotPair>
        {
            new EquipmentSlotPair("main_hand", "greatclub_large"),
            new EquipmentSlotPair("armor", "hide_armor")
        },
        BackpackItemIds = new List<string> { "sack", "rocks" },
        AIBehavior = NPCAIBehavior.RangedThenMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Brute,
        SpriteColor = new Color(0.6f, 0.55f, 0.45f, 1f),
        PanelColor = new Color(0.25f, 0.2f, 0.15f, 0.88f),
        NameColor = new Color(0.9f, 0.85f, 0.75f),
    }
}

```

```

        Description = "Monster Manual hill giant. Large brutish giant with rock
        throwing and devastating club attacks."
    });
}

```

Dependencies:

- Spell resistance system
- Spell-like abilities (darkness, faerie fire)
- Rock throwing (RangedAttackDefinition)
- Light blindness mechanics

Testing:

- Drow SR applies vs spells
- Hill Giant rock throwing works at range
- Equipment loads correctly

Estimated Time: 8 hours

Phase 4 Success Criteria

- [] 3 swarm variants compile and spawn
- [] 8 class-leveled humanoids compile and spawn
- [] Coverage reaches 100% (74 of 74 creatures)
- [] Swarm distraction effect triggers
- [] Drow spell resistance blocks appropriate spells
- [] Hill Giant rock throwing hits at 100 ft range
- [] All encounter tables resolve to valid creature IDs

Phase 4 Effort Summary:

- **Total creatures:** 6 (3 swarms + 3 humanoids, counts vary based on interpretation)
 - **Estimated time:** 2-3 days (16-24 hours)
 - **Files modified:** NPCDatabase_C/D/E/H/L/S.cs (6 files)
 - **New systems required:**
 - Rock throwing ranged attack
 - Light blindness condition
 - **Dependencies:** Swarm mechanics, spell resistance system, equipment
-

Technical Architecture

Files Modified Summary

File	Creatures Added	Phase
NPCDatabase_A.cs	Ankheg	2
NPCDatabase_B.cs	Basilisk, Black Pudding	2
NPCDatabase_C.cs	Choker, Carrion Crawler, Chuul, Centipede Swarm	2
NPCDatabase_D.cs	Darkmantle, Destrachan, Displacer Beast, Dire Boar, Derro, Dwarf Warrior, Drow Warrior, Duergar Warrior	1, 2, 4
NPCDatabase_E.cs	Elf Warrior	4
NPCDatabase_G.cs	Ghoul, Ghast, Grick, Grey Ooze, Gauth, Gibbering Mouther, Grimlock, Giant Constrictor, Ghost variants	1, 2, 3
NPCDatabase_H.cs	Hyena, Hydra variants (5), Hill Giant, Halfling Warrior, Hellwasp Swarm	1, 3, 4
NPCDatabase_K.cs	Kobold Warrior, Krenshar	1
NPCDatabase_L.cs	Locust Swarm	4
NPCDatabase_M.cs	Monitor Lizard, Mind Flayer, Manticore, Mohrg	1, 2
NPCDatabase_O.cs	Otyugh, Owlbear	2
NPCDatabase_P.cs	Phase Spider, Phantom Fungus	2
NPCDatabase_S.cs	Skum, Svirfneblin Warrior, Stone Giant	1, 4
NPCDatabase_V.cs	Violet Fungus, Vampire Spawn	2
NPCDatabase_W.cs	Wight	2

Total Files Modified: 16 NPCDatabase_*.cs files

New Files Created

File	Purpose	Phase
HydraFactory.cs	Parametric hydra generation with variable heads	3
GhostTemplate.cs	Ghost incorporeal undead template	3

Total New Files: 2

Helper Systems Needed

System	Purpose	Implementation Location	Phase
Gaze Attack System	Basilisk, Bodak gaze attacks	NaturalAttackDefinition or new GazeAttackDefinition	2
Stench Aura	Ghast, Troglydte	SpecialAbilities or new AuraEffect	2
Eye Ray Rotation	Gauth multi-ray attacks	SpecialAbilities or new EyeRaySystem	2
Energy Drain	Wight, Vampire Spawn level drain	NaturalAttackDefinition.HasEnergyDrain	2
Multi-Head System	Hydra head count → attacks/HP	HydraFactory parametric generation	3
Incorporeal Mechanics	Ghost 50% miss chance	IsIncorporeal flag + combat system	3
Ability Score Damage	Ghost corrupting touch	NaturalAttackDefinition.AbilityDamage	3
Rock Throwing	Giant ranged attacks	RangedAttackDefinition	4
Light Blindness	Drow, Kobold bright light penalty	SpecialAbilities or condition system	4

Total New Systems: 9 (some may already exist)

Integration Points with DungeonEncounterSpawner

All creatures integrate seamlessly with the existing spawner:

```
// Example: Spawn a custom Drow encounter
var encounter = new EncounterDefinition("Drow Patrol")
    .AddCreatureWithClass("drow_warrior", "Ranger", 3, count: 1) // Leader
    .AddCreature("drow_warrior", count: 3); // Warriors

var result = DungeonEncounterSpawner.PrepareEncounter(encounter);
// result.EnemyIds → ready for SetupEnemyEncounter
```

Template Application:

```
// Ghost template applied at spawn time
var encounter = new EncounterDefinition("Haunted Crypt")
    .AddCreatureWithTemplate("human_fighter_3", "ghost", count: 1)
    .AddCreature("skeleton_warrior", count: 4);

var result = DungeonEncounterSpawner.PrepareEncounter(encounter);
```

Hydra Variants:

```
// Hydra with dynamic head count
var encounter = new EncounterDefinition("Hydra Lair")
    .AddCreature("hydra_7head", count: 1);
```

Code Structure Examples

Example 1: Simple Creature Addition (Hyena)

File: NPCDatabase_H.cs


```

/// <summary>
/// Hyena (CR 1) – Medium animal.
/// MM 3.5e p.274. Pack hunter with trip attack.
/// </summary>
private static void RegisterHyena()
{
    Register(new NPCDefinition
    {
        Id = "hyena",
        Name = "Hyena",
        ChallengeRating = "1",
        Level = 2,
        CharacterClass = "Warrior",
        CreatureType = "Animal",
        HitDice = 2,
        SizeCategory = SizeCategory.Medium,
        IsTallCreature = false,
        STR = 14, DEX = 15, CON = 15, WIS = 13, INT = 2, CHA = 6,
        BAB = 1,
        NaturalArmorBonus = 2,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Bite", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.StrengthOneAndHalf,
                Range = 1, IsPrimary = true,
                TripOnHit = true
            }
        },
        BaseSpeed = 10,
        BaseHitDieHP = 13,
        CreatureTags = new List<string> { "Animal", "MM35" },
        Feats = new List<string> { "Alertness" },
        HasScent = true,
        SpecialAbilities = new List<string>
        {
            "Low-light vision",
            "Scent",
            "Trip: On successful bite, can attempt free trip (no AoO)",
            "Skills: Hide +3, Listen +6, Spot +4"
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIProfileArchetype.Animal,
        SpriteColor = new Color(0.75f, 0.65f, 0.45f, 1f),
        PanelColor = new Color(0.25f, 0.2f, 0.1f, 0.85f),
        NameColor = new Color(0.95f, 0.88f, 0.7f),
        Description = "Monster Manual hyena. Medium pack hunter with trip attack and
scent tracking."
    });
}

```

Then add to RegisterCreatures_H():

```
private static void RegisterCreatures_H()
{
    // ... existing registrations ...
    RegisterHyena();
}
```

Example 2: Lycanthrope Template Application (Custom Werewolf)

File: `LycanthropeFactory.cs`

```
/// <summary>
/// Elf Ranger Werewolf (CR 5) – Natural lycanthrope.
/// Custom variant: Elf ranger 3 + wolf hybrid form.
/// </summary>
public static NPCDefinition ElfRangerWerewolf()
{
    // Step 1: Get base creatures
    NPCDefinition baseElf = NPCDatabase.Get("elf_ranger_3");
    if (baseElf == null)
    {
        Debug.LogError("[LycanthropeFactory] Cannot find elf_ranger_3 base creature.");
    }
    ;
    return null;

    NPCDefinition wolfBase = NPCDatabase.Get("wolf");
    if (wolfBase == null)
    {
        Debug.LogError("[LycanthropeFactory] Cannot find wolf base creature.");
        return null;
    }

    // Step 2: Clone to avoid modifying originals
    NPCDefinition clonedElf = baseElf.Clone();
    NPCDefinition clonedWolf = wolfBase.Clone();

    // Step 3: Apply lycanthrope template
    NPCDefinition lycanthrope = LycanthropeTemplate.Apply(
        baseHumanoid: clonedElf,
        animalBase: clonedWolf,
        lycanthropeId: "elf_ranger_werewolf",
        lycanthropeName: "Elf Ranger Werewolf",
        isAfflicted: false,
        challengeRating: "5"
    );

    // Step 4: Custom adjustments
    lycanthrope.Description = "Natural lycanthrope. Elf ranger merged with wolf, retaining archery skills and wilderness cunning.";
    lycanthrope.SpecialAbilities.Add("Favored Enemy (humanoid): +2 bonus vs one humanoid type");
    lycanthrope.SpecialAbilities.Add("Track: +2 bonus on Survival checks to track");

    Debug.Log("[LycanthropeFactory] Created Elf Ranger Werewolf (CR 5)");

    return lycanthrope;
}
```

Register in NPCDatabase_Lycanthropes.cs:

```
private static void RegisterCreatures_Lycanthropes()
{
    // ... existing lycanthropes ...
    Register(LycanthropeFactory.ElfRangerWerewolf());

    Debug.Log("[NPCDatabase] Registered 9 lycanthrope template variants.");
}
```

Example 3: Hydra Multi-Head Variant

File: HydraFactory.cs (see Phase 3.2 for full implementation)

Usage in NPCDatabase_H.cs:

```
private static void RegisterCreatures_H()
{
    // ... existing registrations ...

    // Hydras (parametric multi-head generation)
    Register(HydraFactory.Hydra5Head());
    Register(HydraFactory.Hydra6Head());
    Register(HydraFactory.Hydra7Head());
    Register(HydraFactory.Hydra8Head());
    Register(HydraFactory.Pyrohydra7Head());

    Debug.Log("[NPCDatabase] Registered 5 hydra variants.");
}
```

Dynamic Hydra Creation (if needed for custom encounters):

```
// In encounter setup code:
NPCDefinition customHydra = HydraFactory.CreateHydra(headCount: 10);
NPCDatabase.Register(customHydra); // Temporary registration
```

Example 4: Size Scaling for Vermin (Swarm)

File: NPCDatabase_L.cs

```

/// <summary>
/// Locust Swarm (CR 3) – Diminutive vermin swarm.
/// MM 3.5e p.237. Dense cloud of biting insects.
/// </summary>
private static void RegisterLocustSwarm()
{
    Register(new NPCDefinition
    {
        Id = "locust_swarm",
        Name = "Locust Swarm",
        ChallengeRating = "3",
        Level = 6,
        CharacterClass = "Warrior",
        CreatureType = "Vermin",
        HitDice = 6,
        SizeCategory = SizeCategory.Diminutive, // Swarm occupies 10 ft square
        IsTallCreature = false,
        STR = 1, DEX = 17, CON = 10, WIS = 10, INT = 0, CHA = 2,
        BAB = 4,
        NaturalArmorBonus = 0,
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Swarm", DamageDice = 6, DamageCount = 2, Count = 1,
                BonusDamageSource = DamageBonusSource.None,
                Range = 0, IsPrimary = true,
                IsSwarmAttack = true // Flag for swarm damage to all in space
            }
        },
        BaseSpeed = 4,
        BaseHitDieHP = 27,
        CreatureTags = new List<string> { "Vermin", "Swarm", "MM35" },
        Feats = new List<string>(),
        HasDarkvision = true,
        SpecialAbilities = new List<string>
        {
            "Swarm Traits: Half damage from slashing/piercing, immune to weapon damage, grappling, tripping. Takes 1.5x damage from area effects.",
            "Distraction (Ex): Living creature in swarm must make Fort DC 12 or nauseated for 1 round",
            "Swarm Attack: 2d6 damage to all creatures in swarm's 10 ft space",
            "Immune to mind effects (mindless)",
            "Darkvision 60 ft."
        },
        EquipmentIds = new List<EquipmentSlotPair>(),
        BackpackItemIds = new List<string>(),
        AIBehavior = NPCAIBehavior.AggressiveMelee,
        AIProfileArchetype = NPCAIPProfileArchetype.Mindless,
        SpriteColor = new Color(0.65f, 0.55f, 0.3f, 0.8f),
        PanelColor = new Color(0.25f, 0.2f, 0.1f, 0.75f),
        NameColor = new Color(0.9f, 0.8f, 0.5f),
        Description = "Monster Manual locust swarm. Cloud of biting insects that engulfs and nauseates victims."
    });
}

```

Swarm Comparison (existing Bat Swarm):

```
// Reference existing swarm for consistency
private static void RegisterBatSwarm()
{
    Register(new NPCDefinition
    {
        Id = "bat_swarm",
        Name = "Bat Swarm",
        ChallengeRating = "2",
        // ... similar swarm structure ...
        NaturalAttacks = new List<NaturalAttackDefinition>
        {
            new NaturalAttackDefinition
            {
                Name = "Swarm", DamageDice = 6, DamageCount = 1, Count = 1,
                BonusDamageSource = DamageBonusSource.None,
                Range = 0, IsPrimary = true,
                IsSwarmAttack = true
            }
        },
        // ... rest of definition ...
    });
}
```

Testing Strategy

Phase 1 Testing (Quick Wins)

Manual Spawn Tests:

1. Spawn 5 random Phase 1 creatures via debug console
2. Verify stats match MM 3.5e values
3. Check equipment loads correctly
4. Confirm AI behavior (melee vs ranged)

Automated Checks:

```
[Test]
public void Phase1_AllCreaturesRegistered()
{
    NPCDatabase.Init();

    string[] phase1Ids = {
        "hyena", "kobold_warrior", "grimlock", "skum", "derro",
        "human_warrior_1", "human_warrior_2", "human_warrior_3",
        // ... all Phase 1 creature IDs
    };

    foreach (string id in phase1Ids)
    {
        Assert.IsNotNull(NPCDatabase.Get(id), $"Creature {id} not registered");
    }
}
```

Phase 2 Testing (Core Creatures)

Combat Ability Tests:

1. **Ghoul Paralysis:** Spawn ghoul, attack player, verify paralysis triggers

2. **Ghast Stench:** Verify 10 ft aura causes sickness
3. **Basilisk Gaze:** Check 30 ft gaze triggers Fort save
4. **Gauth Eye Rays:** Verify 1 ray fires per round, 6 different effects
5. **Darkmantle Darkness:** Confirm darkness spell-like ability works

Spawn Integration Tests:

```
[Test]
public void Phase2_EncounterSpawnerIntegration()
{
    var encounter = new EncounterDefinition("Aberration Test")
        .AddCreature("darkmantle", count: 2)
        .AddCreature("gauth", count: 1);

    var result = DungeonEncounterSpawner.PrepareEncounter(encounter);

    Assert.IsTrue(result.IsValid, "Encounter preparation failed");
    Assert.AreEqual(3, result.Count, "Wrong creature count");
}
```

Phase 3 Testing (Templates)

Template Application Tests:

1. **Hydra Multi-Head:** Verify 5-head hydra has 5 bite attacks
2. **Lycanthrope DR:** Check DR 10/silver applies to custom lycanthropes
3. **Ghost Incorporeal:** Verify 50% miss chance, touch attacks work
4. **Fast Healing:** Confirm hydra regains HP each round

Template Composition Test:

```
[Test]
public void Phase3_GhostTemplateApplication()
{
    NPCDefinition baseFighter = NPCDatabase.Get("human_fighter_3").Clone();
    baseFighter.AppliedTemplateIds = new List<string> { "ghost" };

    NPCDefinition ghost = CreatureTemplateRegistry.ApplyTemplatesClone(baseFighter);

    Assert.AreEqual("Undead", ghost.CreatureType);
    Assert.IsTrue(ghost.IsIncorporeal);
    Assert.AreEqual(0, ghost.NaturalArmorBonus);
    Assert.IsTrue(ghost.SpecialAbilities.Any(s => s.Contains("Horrific Appearance")));
}
```

Phase 4 Testing (Advanced Features)

Size Scaling Tests:

1. Spawn swarms, verify damage applies to all creatures in space
2. Check swarm traits (half damage from slashing/piercing)

Class-Leveled NPC Tests:

1. **Drow SR:** Cast spell at drow, verify SR check
2. **Hill Giant Rock Throw:** Verify 100 ft ranged attack
3. **Spell-Like Abilities:** Drow casts darkness

Coverage Test:

```
[Test]
public void Phase4_FullCoverage()
{
    NPCDatabase.Init();

    // All 74 missing creatures from audit
    string[] allMissingIds = {
        // Phase 1 (30)
        "hyena", "kobold_warrior", /* ... */,
        // Phase 2 (25)
        "darkmantle", "gauth", /* ... */,
        // Phase 3 (13)
        "hydra_5head", "ghost_fighter_3", /* ... */,
        // Phase 4 (6)
        "locust_swarm", "hill_giant", /* ... */
    };

    int foundCount = 0;
    foreach (string id in allMissingIds)
    {
        if (NPCDatabase.Get(id) != null)
            foundCount++;
    }

    Assert.AreEqual(74, foundCount, "Not all missing creatures implemented");
}
```

Success Metrics

Phase-by-Phase Coverage Goals

Phase	Creatures Ad- ded	Cumulative Total	Coverage %	Milestone
Start	0	124	62.6%	Baseline
Phase 1	30	154	77.8%	Quick wins com- plete
Phase 2	25	179	90.4%	Core creatures complete
Phase 3	13	192	96.9%	Templates com- plete
Phase 4	6	198	100%	Full coverage achieved

Quality Metrics

- [] All 74 creatures compile without errors
- [] 100% of encounter table IDs resolve to valid NPCDefinitions

- [] Zero regression in existing 124 creatures
- [] DungeonEncounterSpawner integrates all new creatures
- [] All special abilities documented in SpecialAbilities list
- [] AI behavior appropriate for creature type

Performance Metrics

- [] NPCDatabase.Init() time < 500ms (with all 198 creatures)
- [] Creature lookup (NPCDatabase.Get) < 1ms
- [] Template application < 50ms per creature
- [] Encounter preparation (20 creatures) < 200ms

Documentation Metrics

- [] Each creature has XML doc comment with MM page reference
- [] All special abilities listed in SpecialAbilities
- [] CreatureTags include “MM35” for all Monster Manual creatures
- [] Description field summarizes creature’s tactical role

Appendix: Complexity Breakdown

By Implementation Difficulty

Complexity	Count	Examples	Avg Time per Creature
Easy	25	Hyena, Kobold, Grimlock, Owlbear, basic animals	15-30 min
Medium	35	Darkmantle, Ghoul, Otyugh, Basilisk, Hill Giant	30-60 min
Hard	14	Gauth, Mind Flayer, Hydras, Ghost, Lycanthropes	1-3 hours
TOTAL	74	—	~14-20 days

By Creature Type

Type	Count	Examples
Aberrations	10	Darkmantle, Choker, Grick, Gauth, Mind Flayer
Animals	5	Hyena, Monitor Lizard, Dire Boar, Giant Constrictor
Giants	3	Hill Giant, Stone Giant, Ettin
Humanoids	14	Kobold, Grimlock, Drow, Duer-gar, racial warriors
Magical Beasts	11	Owlbear, Ankheg, Basilisk, Hydras (5)
Oozes	2	Grey Ooze, Black Pudding
Outsiders	0	(Future phases: Devils, Demons, Celestials)
Plants	2	Violet Fungus, Phantom Fungus
Undead	9	Ghoul, Ghast, Wight, Vampire Spawn, Ghost (4)
Vermin (Swarms)	3	Locust Swarm, Centipede Swarm, Hellwasp Swarm
Templates	15	Lycanthropes (3 custom), Hydras (5), Ghosts (4)

By Special Ability Type

Ability	Count	Implementation Notes
Paralysis	5	HasParalysisOnHit flag
Poison	3	HasPoisonOnHit, PoisonType
Disease	2	HasDiseaseOnHit, Disease-Type
Energy Drain	3	HasEnergyDrain, DrainLevels
Gaze Attacks	2	New GazeAttackDefinition
Spell-Like Abilities	6	SpellLikeAbilityDefinition
Spell Resistance	3	SpellResistance property
Incorporeal	4	IsIncorporeal flag
Fast Healing	5	HasFastHealing, FastHealingRate
Swarm	3	IsSwarmAttack flag
Rock Throwing	2	RangedAttackDefinition
Improved Grab	8	Existing NaturalAttackDefinition
Trip	2	TripOnHit flag

Implementation Roadmap Timeline

Week 1: Phases 1-2 (Foundation)

- **Days 1-2:** Phase 1 (30 quick wins) — 77.8% coverage
- **Days 3-6:** Phase 2 (25 core creatures) — 90.4% coverage
- **Day 7:** Testing, bug fixes, integration validation

Week 2: Phases 3-4 (Advanced)

- **Days 1-4:** Phase 3 (template systems, 13 variants) — 96.9% coverage
- **Days 5-6:** Phase 4 (final 6 creatures) — 100% coverage
- **Day 7:** Final testing, documentation, polish

Total Project Duration: 14-16 days (120-140 hours)

Conclusion

This implementation plan provides a clear, phased approach to adding 74 missing creatures to the D&D 3.5e NPCDatabase. By organizing work into 4 progressive phases — Quick Wins, Core Creatures, Templates, and Advanced Features — the project maintains momentum while building increasingly complex systems.

Key Success Factors:

1. Reuse existing frameworks (templates, spawner, class engine)
2. Progressive complexity (easy → hard)
3. Clear testing criteria at each phase
4. Consistent coding style and documentation
5. Minimal new systems (9 total, many optional)

Final Coverage: 198/198 creatures (100%)

Expected Effort: 14-20 days

Risk Level: Low (incremental, testable, reversible)

The phased approach allows for early validation and course correction, ensuring high-quality implementations that integrate seamlessly with the existing combat and encounter systems.

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